

2017 Final Sorusu:

$a > 0$ olmak üzere $6 + \int_a^x \frac{f(t)}{t^2} dt = 2\sqrt{x}$ eşitliğini sağlayan

f fonksiyonu ve a sayısını bulunuz.

$$6 + \int_a^x \frac{f(t)}{t^2} dt = 2\sqrt{x} \quad \xrightarrow{\text{Törev}} \quad \frac{f(x)}{x^2} = \frac{1}{\sqrt{x}} \Rightarrow \boxed{f(x) = x^{3/2}}$$

$\downarrow x=a$ yazarsak

$$6 + \int_a^a \frac{f(t)}{t^2} dt = 2\sqrt{a} \quad \Rightarrow \quad 3 = \sqrt{a} \quad \Rightarrow \quad \boxed{a=9}$$

⊗ 2017 Bütünleme Sorusu

$$\int \sec^2 \sqrt{x} dx = ?$$

$$\sqrt{x} = t \quad \frac{1}{2\sqrt{x}} dx = dt \Rightarrow dx = 2\sqrt{x} dt = 2t dt$$

$$\int \sec^2 \sqrt{x} dx = 2 \int t \sec^2 t dt \quad \begin{cases} t=u & dt=du \\ \sec^2 t dt = dv & v = \tan t \end{cases}$$

$$= 2 \left[t \tan t - \int \tan t dt \right] = 2t \tan t + 2 \ln |\cos t| + c$$

$$= 2\sqrt{x} \tan \sqrt{x} + 2 \ln |\cos \sqrt{x}| + c$$

2017 2. Vize

$$I = \int_0^{\pi/4} e^{\tan x - 2 \ln |\cos x|} dx = ?$$

$$e^{\tan x} \cdot \frac{e^{\ln(\cos x)^2}}{\frac{1}{\cos^2 x}} = \frac{e^{\tan x}}{\cos^2 x}$$

$$\Rightarrow I = \int_0^{\pi/4} \frac{e^{\tan x}}{\cos^2 x} dx$$

$$\tan x = u$$

$$\frac{dx}{\cos^2 x} = du$$

$$x = \frac{\pi}{4} \rightarrow u = 1$$

$$x = 0 \rightarrow u = 0$$

$$= \int_0^1 e^u du = e^u \Big|_0^1 = \underline{\underline{e-1}}$$

2017 Mazeret Sınavı

$$x \geq 1, f(x) = \int_2^{x+1} (t-1)^t dt \Rightarrow f''(1) = ?$$

↓ Türev

$$f'(x) = (x+1-1)^{x+1} = x^{x+1}$$

$$\rightarrow f'(1) = 1^2 = 1$$

↓ logaritmik türev

$$\ln f'(x) = (x+1) \ln x$$

↓ Türev

$$\frac{f''(x)}{f'(x)} = \ln x + \frac{x+1}{x} \Rightarrow \frac{f''(1)}{1} = \ln 1 + \frac{2}{1} \Rightarrow \boxed{f''(1) = 2}$$

2017 - Mazeret Sınavı

$$\int_1^e \frac{\ln x^2}{x(1+\ln^2 x)} = \int_1^e \frac{2 \ln x}{x(1+\ln^2 x)} dx = \int_0^1 \frac{2u}{1+u^2} du = \ln |1+u^2| \Big|_0^1 = \underline{\ln 2}$$

$$\ln x = u \quad \frac{dx}{x} = du$$

$$x=e \rightarrow u=1$$

$$x=1 \rightarrow u=0$$

2017- Final

* $f(x) = x\sqrt{16-x^2}$ fonksiyonunun yerel ve mutlak ekstremumlarını bulunuz.

$$D(f) = [-4, 4]$$

$$f'(x) = \sqrt{16-x^2} + \frac{x \cdot -2x}{2\sqrt{16-x^2}} = \frac{16-2x^2}{\sqrt{16-x^2}} \rightarrow F' = 0 \rightarrow x^2 = 8 \rightarrow \boxed{x = \pm 2\sqrt{2}} \text{ K.N.}$$

$$x = \pm 4 \text{ Uç Nokta} \rightarrow f' \text{ tanımsız} \rightarrow \boxed{x = \pm 4} \text{ K.N.}$$

x	-4	-2√2	2√2	4
f'	—	+	—	—
f	↘	↗	↘	↘
	Y.max	Y.min	Y.max	Y.min

$$f(-4) = f(4) = 0$$

$$f(2\sqrt{2}) = 8$$

$$f(-2\sqrt{2}) = -8$$

Yerel Max
(-4, 0), (2√2, 8)

Yerel Min
(-2√2, -8), (4, 0)

Mutlak Max
(2√2, 8)

Mutlak Min
(-2√2, -8)

- * $f(x) = \frac{x^5 - 3}{x^4}$ eğrisinin asimptotlarını, artan / azalan olduğu aralıkları, konvoksliğini, ekstremleri kesim noktalarını araştırıp eğriyi çiziniz.

① $x=0$ Dikey asimptot

② $f(x) = \frac{x^5 - 3}{x^4} = \boxed{x} - \frac{3}{x^4} \Rightarrow y=x$ eğik asimptot (veya $\frac{x^5 - 3}{-x^5} \cdot \frac{x^4}{\boxed{x}} \rightarrow y=x$ c.A.)

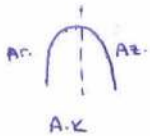
③ $f'(x) = \frac{5x^4 \cdot x^4 - 4x^3(x^5 - 3)}{x^8} = \frac{x^3(x^5 + 12)}{x^8} = \frac{x^5 + 12}{x^5}$
 $f' = 0 \Rightarrow x^5 = -12 \Rightarrow \boxed{x = -\sqrt[5]{12}}$ K.N.
 f' tanımsız $\boxed{x=0}$ K.N. değil (f tanımsız)

④ $f''(x) = \frac{5x^4 \cdot x^5 - 5x^4(x^5 + 12)}{x^{10}} = -\frac{60}{x^6}$
 $f'' = 0 \Rightarrow \text{çok yok}$
 f'' tanımsız $\boxed{x=0}$ A.K.K.
 Büküm Nok. olamaz (f tanımsız)

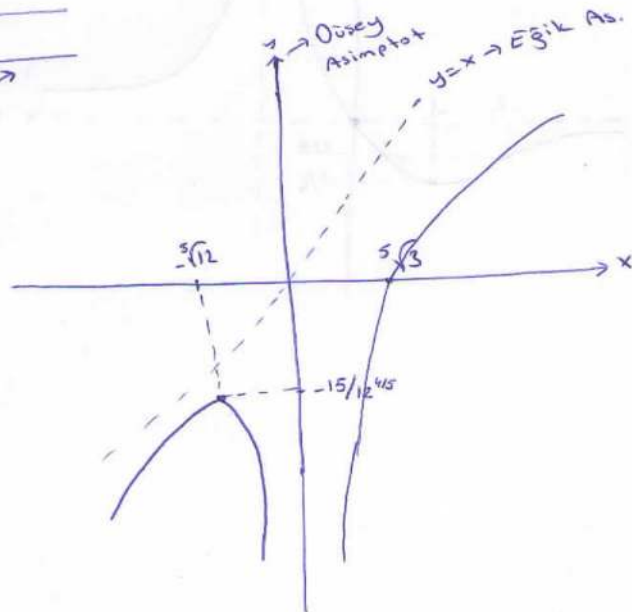
⑤ $y=0 \Rightarrow x^5 - 3 = 0 \Rightarrow \boxed{x = \sqrt[5]{3}}$

⑥

x	$-\infty$	$-\sqrt[5]{12}$	0	∞
f'	+	0	-	+
f''	-	-	-	-
f	↗	↘	↘	↗



$x = -\sqrt[5]{12} \Rightarrow y = \frac{-12 - 3}{(-\sqrt[5]{12})^4} = -\frac{15}{12^{4/5}}$



* $y = \frac{x}{(x-1)^2}$ eğrisini çiziniz.

① $x=1$ dikey As. ② $\lim_{x \rightarrow \pm\infty} \frac{x}{(x-1)^2} = 0$ olduğundan $y=0$ Yatay As.

$$y=0 \Rightarrow x=0$$

(0,0) da yatay asimptot ve eğri kesişir

③ $y' = \frac{(x-1)^2 - 2(x-1) \cdot x}{(x-1)^4} = -\frac{x+1}{(x-1)^3}$

$y'=0 \Rightarrow x=-1$ K.N.
 y' tanımsız $x=1$ K.N. olamaz (fonk. tanımsız)

④ $y'' = -\frac{(x-1)^2 - 3(x-1) \cdot (x+1)}{(x-1)^6} = \frac{2x+4}{(x-1)^4}$

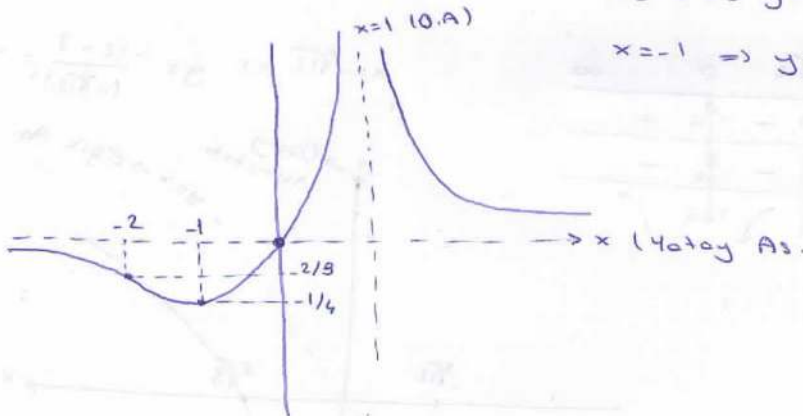
$y''=0 \Rightarrow x=-2$
 y'' tanımsız $x=1$ G.K.K. (B.N. olamaz)

x	$-\infty$	-2	-1	1	∞
y'	-	-	0	+	-
y''	-	+	+	+	+
y	$\searrow$	$\nearrow$	$\nearrow$	$\searrow$	



$$x=-2 \Rightarrow y = -\frac{2}{9}$$

$$x=-1 \Rightarrow y = -\frac{1}{4}$$



2017 Mazeret

OOT kullanarak her $x > 0$ için $\frac{x}{1+x} < \ln(1+x) < x$ olduğunu gösterin.

$f(x) = \ln(1+x)$, olsun. $[0, x]$ aralığı için:

① $f(x)$, $[0, x]$ de sürekli ② $f'(x) = \frac{1}{1+x}$, $(0, x)$ de tanımlıdır.

O.O.T. şartları sağlanır ✓

0 halde:

$$f'(c) = \frac{f(x) - f(0)}{x - 0} \quad \text{olacak şekilde bir } c \in (0, x) \text{ vardır.}$$

$$\begin{aligned} \downarrow \\ \frac{1}{1+c} &= \frac{\ln(1+x)}{x} \rightarrow c > 0 \Rightarrow \frac{1}{1+c} = \frac{\ln(1+x)}{x} < 1 \\ &\downarrow \\ &\boxed{\ln(1+x) < x} \\ &\rightarrow c < x \Rightarrow \frac{1}{1+c} = \frac{\ln(1+x)}{x} > \frac{1}{1+x} \\ &\downarrow \\ &\boxed{\ln(1+x) > \frac{x}{1+x}} \end{aligned}$$

$$\begin{aligned} (*) \int x^3 e^{x^2} dx &= \int x \cdot \underbrace{x^2}_{\frac{du}{2}} \cdot \underbrace{e^{x^2}}_{e^u} dx = \frac{1}{2} \int u e^u du = \frac{1}{2} (u e^u - \int e^u du) \\ x^2 &= u \quad 2x dx = du \\ u &= t \quad du = dt \\ e^u du &= dv \quad v = e^u \quad \left| \begin{aligned} &= \frac{1}{2} (u e^u - e^u) + C \\ &= \frac{1}{2} x^2 e^{x^2} - \frac{e^{x^2}}{2} + C \end{aligned} \right. \end{aligned}$$

2016 - Mazeret

$$f(x) = (1 + \ln x)^x, \quad g(x) = \int_x^{f(x)} \cos^2 x \, dx \Rightarrow f'(1) = ? \quad g'(1) = ? \quad (x > 0 \text{ için})$$

$$f(x) = (1 + \ln x)^x \xrightarrow{\text{L. Türev}} \ln f(x) = x \ln(1 + \ln x) \xrightarrow{\text{Türev}} \frac{f'(x)}{f(x)} = \ln(1 + \ln x) + x \cdot \frac{1}{1 + \ln x}$$

$$g(x) = \int_x^{f(x)} \cos^2 x \, dx \quad \downarrow \text{Türev}$$

$$\begin{aligned} \downarrow x=1 \\ \frac{f'(1)}{\frac{f(1)}{1}} &= \frac{0}{1} + \frac{1}{1 + \ln 1} \\ &\downarrow \\ &\boxed{f'(1) = 1} \end{aligned}$$

$$\begin{aligned} g'(x) &= f'(x) \cdot \cos^2 f(x) - \cos^2 x \xrightarrow{x=1} g'(1) = \frac{f'(1)}{1} \cdot \cos^2 f(1) - \cos^2 f(1) \\ &\boxed{g'(1) = 0} \end{aligned}$$

⊗ $\int \cot^n x dx$ integrali için indirgeme formülü bulup

$\int \cot^3 x dx$ integralini bu formülle hesaplayın.

$$\operatorname{cosec}^2 x = 1 + \cot^2 x$$

$$I_n = \int \cot^n x dx = \int (\cot^n x)^{n-2} \cdot \frac{\cot^2 x dx}{(\operatorname{cosec}^2 x - 1)}$$

$$= \int \underbrace{(\cot^{n-2} x)}_{u^{n-2}} \cdot \underbrace{\operatorname{cosec}^2 x dx}_{-du} - \int \underbrace{(\cot^{n-2} x) dx}_{I_{n-2}}$$

$$I_n = - \frac{u^{n-1}}{n-1} - I_{n-2}$$

$$\boxed{I_n = - \frac{(\cot x)^{n-1}}{n-1} - I_{n-2}} \Rightarrow I_3 = - \frac{\cot^2 x}{2} - \int \cot x dx$$

$$= - \frac{\cot^2 x}{2} - \ln |\sin x| + C$$

2017 Final Sorusu:

$$I = \int \cos(\ln x) dx = ?$$

↓ Kısmi int.

$$\cos(\ln x) = u \quad dx = dv$$

$$-\frac{1}{x} \sin(\ln x) dx = dv \quad v = x$$

$$\sin(\ln x) = u \quad dx = dv$$

$$\frac{1}{x} \cos(\ln x) dx = dv \quad v = x$$

$$I = x \cos(\ln x) + \int \sin(\ln x) dx$$

$$= x \cos(\ln x) + x \sin(\ln x) - \underbrace{\int \cos(\ln x) dx}_I$$

$$\Downarrow$$

$$2I = x \cos(\ln x) + x \sin(\ln x)$$

$$I = \frac{1}{2} (x \cos(\ln x) + x \sin(\ln x)) + C$$

$$\begin{aligned}
 (*) \quad \int_0^{\pi/2} \frac{\sin 2x}{e^{\sin^2 x} + e^{-\sin^2 x}} dx &= \int_0^1 \frac{du}{e^u + e^{-u}} = \int_0^1 \frac{e^u du}{e^{2u} + 1} = \int_1^e \frac{dt}{t^2 + 1} = \text{Arctan } t \Big|_1^e \\
 &= \text{Arctan } e - \text{Arctan } 1 \\
 &= \text{Arctan } e - \frac{\pi}{4}
 \end{aligned}$$

$\sin^2 x = u$
 $2 \sin x \cos x dx = du$
 $x = \frac{\pi}{2} \Rightarrow u = 1$
 $x = 0 \Rightarrow u = 0$

$$\left. \begin{aligned} e^u &= t \\ e^u du &= dt \\ u=1 &\Rightarrow t=e \end{aligned} \right\}$$

$$\begin{aligned}
 (*) \quad \int \frac{e^{1/x}}{1-3x+3x^2-x^3} dx &=? \\
 1-3x+3x^2-x^3 &= (1-x)^3 \\
 \bullet \quad \frac{1}{1-x} = u &\Rightarrow \frac{dx}{(1-x)^2} = du
 \end{aligned}$$

$$\left. \begin{aligned} u=t \quad du=dt \\ e^u du = dv \quad v=e^u \end{aligned} \right\} \quad I = \int \frac{\widehat{e^{1/x}}}{\frac{1}{u} \cdot (1-x)^2} \frac{dx}{(1-x)^2} = \int u e^u du$$

Kismi int.

$$\begin{aligned}
 \Rightarrow \quad &= u e^u - \int e^u du = u e^u - e^u + C \\
 &= \frac{1}{1-x} \cdot e^{1/x} - e^{1/x} + C
 \end{aligned}$$

$$\begin{aligned}
 (*) \quad \int \frac{dx}{\tan^5 x} &= \int \frac{\cos^5 x}{\sin^5 x} dx = \int \frac{(\cos^2 x)^2 \cdot \cos x}{\sin^5 x} dx \quad \begin{aligned} \sin x &= u \\ \cos x dx &= du \end{aligned} \\
 &= \int \frac{(1-u^2)^2}{u^5} du = \int (u^{-5} - 2u^{-3} - u^{-1}) du \\
 &= \frac{u^{-4}}{-4} - \frac{2u^{-2}}{-2} - \ln|u| + C = \\
 &= -\frac{1}{4 \sin^4 x} + \frac{1}{\sin^2 x} - \ln|\sin x| + C
 \end{aligned}$$

$$(*) \int \frac{x+3}{x^2-2x+10} dx = ?$$

$$\begin{aligned} \frac{1}{2} \int \frac{2x+6}{x^2-2x+10} dx &= \frac{1}{2} \int \frac{2x-2+8}{x^2-2x+10} dx = \frac{1}{2} \int \frac{2x-2}{x^2-2x+10} dx + 4 \int \frac{\overset{du}{dx}}{\underset{u}{(x-1)^2+9}} \\ &= \frac{1}{2} \ln |x^2-2x+10| + \frac{4}{3} \operatorname{Arctan} \frac{x-1}{3} + C \end{aligned}$$

$$(*) I = \int \operatorname{Arc} \cos x \, dx = ?$$

$$\begin{aligned} \left. \begin{array}{l} u = \operatorname{Arc} \cos x \quad dv = dx \\ du = -\frac{dx}{\sqrt{1-x^2}} \quad v = x \end{array} \right\} I &= x \operatorname{Arc} \cos x + \int \frac{x}{\sqrt{1-x^2}} dx \Rightarrow \begin{array}{l} 1-x^2 = u \\ -2x dx = du \end{array} \\ &= x \operatorname{Arc} \cos x - \frac{1}{2} \int \frac{du}{\sqrt{u}} \\ &= x \operatorname{Arc} \cos x - \sqrt{u} + C = x \operatorname{Arc} \cos x - \sqrt{1-x^2} + C \end{aligned}$$

$$(*) I = \int e^{2x} \cdot \cos 3x \, dx = ?$$

$$\left\{ \begin{array}{l} u = \cos 3x \quad e^{2x} dx = dv \\ du = -3 \sin 3x \, dx \quad v = \frac{e^{2x}}{2} \end{array} \right. \quad \left\{ \begin{array}{l} u = \sin 3x \quad e^{2x} dx = dv \\ du = 3 \cos 3x \, dx \quad v = \frac{e^{2x}}{2} \end{array} \right.$$

$$I = \frac{e^{2x}}{2} \cos 3x + \frac{1}{2} \int e^{2x} \sin 3x \, dx$$

$$= \frac{e^{2x}}{2} \cos 3x + \frac{3}{2} \left[\frac{e^{2x}}{2} \sin 3x - \frac{3}{2} \underbrace{\int e^{2x} \cos 3x \, dx}_I \right]$$

$$\left(1 + \frac{9}{4}\right) I = \frac{e^{2x}}{2} \cos 3x + \frac{3}{4} e^{2x} \sin 3x$$

$$I = \frac{2}{13} e^{2x} \cos 3x + \frac{3}{13} e^{2x} \sin 3x + C$$

$$(*) \int_0^1 e^{\text{Arcsin} x} dx = ?$$

$$\left. \begin{array}{l} \text{Arcsin} x = u \Rightarrow x = \sin u \\ dx = \cos u du \\ \text{E.S.} \quad \text{Y.S.} \\ x=1 \Rightarrow u=\pi/2 \\ x=0 \Rightarrow u=0 \end{array} \right\} I = \int_0^{\pi/2} e^u \cos u du \rightarrow \begin{array}{l} \text{Kismi} \\ \text{int.} \end{array} \begin{array}{l} \cos u = t \\ -\sin u du = dt \\ e^u du = dv \\ v = e^u \end{array}$$

$$I = e^u \cos u \Big|_0^{\pi/2} + \int_0^{\pi/2} e^u \sin u \quad \begin{array}{l} \text{Kismi} \\ \text{int.} \end{array} \begin{array}{l} \sin u = t \\ e^u du = dv \\ v = e^u \end{array} \cos u du = dt$$

$$I = \underbrace{e^{\pi/2} \cos \frac{\pi}{2}}_0 - \underbrace{e^0 \cos 0}_1 + e^u \sin u \Big|_0^{\pi/2} - \underbrace{\int_0^{\pi/2} e^u \cos u du}_I$$

$$2I = -1 + e^{\pi/2} \sin \frac{\pi}{2} - \frac{e^0 \sin 0}{0} \Rightarrow 2I = e^{\pi/2} - 1 \Rightarrow \boxed{I = \frac{e^{\pi/2} - 1}{2}}$$

$$(*) \int \text{Arctan} \frac{1}{x} dx = x \cdot \text{Arctan} \frac{1}{x} + \int \frac{x}{1+x^2} dx$$

$$\left. \begin{array}{l} \text{Arctan} \frac{1}{x} = u \quad dx = dv \\ -\frac{1}{x^2} dx = du \quad v = x \\ \frac{-\frac{1}{x^2} dx}{1 + \frac{1}{x^2}} = du \end{array} \right\} = x \text{Arctan} \frac{1}{x} + \frac{1}{2} \ln |1+x^2| + C$$

$$-\frac{dx}{1+x^2} = du$$

Hepinize 2. vizelerinizde başarılar dilerim...

Sevgiler,
Pinar Albayrak